

```
In [4]: import pandas as pd
```

```
In [5]: df = pd.read_csv('D:\Desktop\Data Science\python\Messy_Employee_dataset.csv')
```

```
In [6]: df.info
df.head(5)
```

Out[6]:

	Employee_ID	First_Name	Last_Name	Age	Department_Region	Status	Join_Date
0	EMP1000	Bob	Davis	25.0	DevOps-California	Active	4/2/2021
1	EMP1001	Bob	Brown	NaN	Finance-Texas	Active	7/10/2020
2	EMP1002	Alice	Jones	NaN	Admin-Nevada	Pending	12/7/2023
3	EMP1003	Eva	Davis	25.0	Admin-Nevada	Inactive	11/27/2021
4	EMP1004	Frank	Williams	25.0	Cloud Tech-Florida	Active	1/5/2022

```
In [7]: df[df.isnull().any(axis = 1)]
```

Out[7]:

	Employee_ID	First_Name	Last_Name	Age	Department_Region	Status	Join_Date
1	EMP1001	Bob	Brown	NaN	Finance-Texas	Active	7/10/2020
2	EMP1002	Alice	Jones	NaN	Admin-Nevada	Pending	12/7/2023
6	EMP1006	Frank	Jones	NaN	Admin-Nevada	Active	4/3/2020
9	EMP1009	Charlie	Johnson	NaN	DevOps-New York	Active	8/4/2022
11	EMP1011	David	Davis	NaN	DevOps-California	Active	9/4/2023
...	...	...	...	...	...	...	...
1006	EMP2006	Grace	Brown	NaN	HR-Illinois	Inactive	10/25/2023
1009	EMP2009	David	Garcia	NaN	Finance-Illinois	Pending	6/11/2024
1012	EMP2012	Grace	Miller	35.0	DevOps-Florida	Inactive	4/15/2023
1013	EMP2013	Eva	Williams	NaN	HR-Illinois	Pending	2/6/2020
1015	EMP2015	David	Miller	30.0	HR-California	Active	8/19/2023

233 rows × 8 columns

Issues in the data were determined, let's solve them one by one :)

>>>>> Missing data in the age column

```
In [8]: # I didn't just want to fill the missing values with the general mean but has taken
total_mean= df["Age"].mean()
df["Age"] = df["Age"].fillna( df.groupby("Department_Region")["Age"].transform("mean") )
df.head(5)
```

```
Out[8]:
```

	Employee_ID	First_Name	Last_Name	Age	Department_Region	Status	Join_Date
0	EMP1000	Bob	Davis	25.0	DevOps-California	Active	4/2/2021
1	EMP1001	Bob	Brown	34.0	Finance-Texas	Active	7/10/2020
2	EMP1002	Alice	Jones	32.0	Admin-Nevada	Pending	12/7/2023
3	EMP1003	Eva	Davis	25.0	Admin-Nevada	Inactive	11/27/2021
4	EMP1004	Frank	Williams	25.0	Cloud Tech-Florida	Active	1/5/2022

>>>>> Missing data in the salary column

```
In [9]: total_mean= df["Salary"].mean()
df["Salary"] = df["Salary"].fillna( df.groupby("Department_Region")["Salary"].transform("mean") )
df.head(5)
```

```
Out[9]:
```

	Employee_ID	First_Name	Last_Name	Age	Department_Region	Status	Join_Date
0	EMP1000	Bob	Davis	25.0	DevOps-California	Active	4/2/2021
1	EMP1001	Bob	Brown	34.0	Finance-Texas	Active	7/10/2020
2	EMP1002	Alice	Jones	32.0	Admin-Nevada	Pending	12/7/2023
3	EMP1003	Eva	Davis	25.0	Admin-Nevada	Inactive	11/27/2021
4	EMP1004	Frank	Williams	25.0	Cloud Tech-Florida	Active	1/5/2022

>>>>> incorrect symbol in the number column

```
In [10]: df["Phone"] = df["Phone"].astype(str).str.replace("-", "")
df.head(3)
```

Out[10]:

	Employee_ID	First_Name	Last_Name	Age	Department_Region	Status	Join_Date	
0	EMP1000	Bob	Davis	25.0	DevOps-California	Active	4/2/2021	59
1	EMP1001	Bob	Brown	34.0	Finance-Texas	Active	7/10/2020	65
2	EMP1002	Alice	Jones	32.0	Admin-Nevada	Pending	12/7/2023	88



Let's now do a small test

In [11]: `df[df.isnull().any(axis = 1)]`

Out[11]:

	Employee_ID	First_Name	Last_Name	Age	Department_Region	Status	Join_Date	Salary
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Yaay, Now the data is clean and ready for some analysis!

In [12]: `df[["Age", "Salary"]].corr()`

Out[12]:

	Age	Salary
Age	1.00000	0.07741
Salary	0.07741	1.00000